GLYDE-2 Study Protocol

A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of Once-Daily Oral SYN-4417 as Add-on to Metformin in Adults with Type 2 Diabetes Mellitus Inadequately Controlled on Metformin Monotherapy

Protocol Number: SYN-T2D-301

National Clinical Trial (NCT) Identified Number: Pending

EudraCT / EU CT Number: Pending

Investigational Product: SYN-4417 (oral GLP-1 receptor agonist)

Sponsor: Synthetic Therapeutics Ltd (fictional)

Coordinating Investigator: Dr. Jane Example, MD PhD

Protocol Version 2.0

Date: 14 March 2026

CONFIDENTIAL: This document contains confidential information belonging to the sponsor. It is a synthetic protocol created to test protocol-review software; the drug, sponsor, investigators and data are fictional.

STATEMENT OF COMPLIANCE

The trial will be carried out in accordance with International Council for Harmonisation Good Clinical Practice (ICH GCP E6(R3)), the Declaration of Helsinki, and all applicable national regulations. All personnel involved in the conduct of this study have completed Human Subjects Protection and ICH GCP training.

The protocol, informed consent form(s), recruitment materials, and all participant materials will be submitted to the Institutional Review Board (IRB) or Independent Ethics Committee (IEC) for review and approval. Approval of both the protocol and the consent form must be obtained before any participant is enrolled.

INVESTIGATOR'S SIGNATURE

I have read this protocol and agree to conduct the study as outlined herein, in accordance with ICH GCP and all applicable regulatory requirements.

Principal Investigator: _____ Date: _____

Site name and number: _____

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1 PROTOCOL SUMMARY

1.1 Synopsis

Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of Once-Daily Oral SYN-4417 as Add-on to Metformin in Adults with Type 2 Diabetes Mellitus Inadequately Controlled on Metformin Monotherapy

Brief title: GLYDE-2

Protocol number: SYN-T2D-301

Phase: Phase 3

Study description: This is a 52-week, multicenter, randomized, double-blind, placebo-controlled, parallel-group study of SYN-4417, an oral small-molecule glucagon-like peptide-1 (GLP-1) receptor agonist, given once daily in addition to stable metformin. Participants are randomized in a 1:1:1 ratio to SYN-4417 7 mg, SYN-4417 14 mg, or matching placebo. A 2-week screening period precedes randomization and a 4-week safety follow-up period follows the last dose.

Population: Approximately 600 adults aged 18 to 70 years with type 2 diabetes mellitus (T2DM), HbA1c between 7.5% and 9.5% inclusive, and a stable dose of metformin of at least 1500 mg/day for at least 12 weeks before screening.

Number of sites: Approximately 85 sites in 6 countries (United States, Canada, United Kingdom, Germany, Spain and Poland).

Study intervention: SYN-4417 7 mg, SYN-4417 14 mg, or placebo tablets, taken orally once daily in the morning for 52 weeks. Participants randomized to 14 mg start on 7 mg for 4 weeks before escalating.

Primary endpoint: Change from baseline in HbA1c at Week 26.

Key secondary endpoints: Change from baseline in body weight at Week 26; proportion of participants reaching HbA1c < 7.0% at Week 26.

Study duration: Approximately 58 weeks per participant (2-week screening, 52-week treatment, 4-week follow-up). Total study duration from first participant screened to last participant's last visit is estimated at 30 months.

Masking: Double-blind (participant, investigator, outcomes assessor and sponsor study team).

Sample size: 600 participants (200 per arm) provide 90% power to detect a 0.3% difference in HbA1c change between SYN-4417 14 mg and placebo, assuming a standard deviation of 1.1% and a two-sided alpha of 0.05.

1.2 Schedule of Activities

Procedure	V1	V2	V3	V4	V5	V6	V7	V8
Study week	-2	0	4	12	26	39	52	56
Visit window (days)	+/-3	0	+/-3	+/-5	+/-5	+/-7	+/-7	+/-7
Informed consent	X							
Eligibility review	X	X						
Medical history	X							
Physical examination	X						X	
Vital signs	X	X	X	X	X	X	X	X
Body weight	X	X		X	X		X	
12-lead ECG	X						X	
HbA1c (central lab)	X	X		X			X	
Fasting plasma glucose	X	X	X	X	X	X	X	
Serum chemistry and eGFR	X		X	X	X		X	X
Lipid panel		X					X	
Pregnancy test (WOCBP)	X	X		X	X	X	X	
Randomization (IWRS)		X						
Dispense study drug		X	X	X	X	X		
Drug accountability			X	X	X	X	X	
Adverse event review		X	X	X	X	X	X	X
Concomitant medications	X	X	X	X	X	X	X	X
SMBG diary review		X	X	X	X	X	X	
PRO questionnaires		X			X		X	

Abbreviations: V = visit; WOCBP = women of childbearing potential; IWRS = interactive web response system; SMBG = self-monitoring of blood glucose; PRO = patient-reported outcome. Week 56 (V8) is the safety follow-up visit, 4 weeks after the last dose.

2 INTRODUCTION

2.1 Study Rationale

Type 2 diabetes mellitus affects more than 500 million adults worldwide, and roughly half of treated patients do not reach the glycemic target of HbA1c < 7.0%. Metformin remains first-line therapy, but most patients require a second agent within a few years of diagnosis. Injectable GLP-1 receptor agonists lower HbA1c and body weight, yet the injection burden limits uptake. An oral, small-molecule GLP-1 receptor agonist that does not require fasting administration could widen access to this class.

2.2 Background

SYN-4417 is a non-peptide, orally bioavailable GLP-1 receptor agonist with a half-life of approximately 21 hours, supporting once-daily dosing without food restrictions. In the 12-week Phase 2 study SYN-T2D-201 (N = 312), SYN-4417 14 mg reduced HbA1c by 1.2% from a baseline of 8.1%, compared with 0.3% for placebo, and reduced body weight by 3.4 kg compared with 0.8 kg. The most common adverse events were nausea (18%), diarrhea (9%) and vomiting (5%), mostly mild and occurring during dose escalation.

2.3 Risk/Benefit Assessment

Known potential risks include gastrointestinal adverse events, acute pancreatitis, gallbladder events, and hypoglycemia when combined with insulin secretagogues. Rodent studies showed thyroid C-cell tumours, a class finding of uncertain relevance to humans; participants with a personal or family history of medullary thyroid carcinoma are excluded. Potential benefits include improved glycemic control and weight loss. The sponsor considers the overall benefit/risk balance favourable.

3 OBJECTIVES AND ENDPOINTS

Objective	Endpoint	Time frame
Primary: demonstrate superiority of SYN-4417 14 mg over placebo in glycemic control	Change from baseline in HbA1c (%)	Week 26
Key secondary: effect on body weight	Change from baseline in body weight (kg)	Week 26
Key secondary: glycemic target attainment	Proportion of participants with HbA1c < 7.0%	Week 26
Secondary: fasting glucose	Change from baseline in fasting plasma glucose (mg/dL)	Weeks 26 and 52
Secondary: durability	Change from baseline in HbA1c (%)	Week 52
Safety	Incidence of treatment-emergent adverse events, including gastrointestinal events and hypoglycemia	Week 0 to Week 56
Exploratory: patient-reported outcomes	Change in DTSQ and IWQOL-Lite-CT total scores	Week 52

The primary estimand compares the mean change in HbA1c from baseline to Week 26 between SYN-4417 14 mg and placebo in the randomized population.

4 STUDY DESIGN

4.1 Overall Design

This is a Phase 3, multicenter, randomized, double-blind, placebo-controlled, parallel-group study with three arms. Eligible participants are randomized 1:1:1 by IWRS, stratified by baseline HbA1c (<= 8.5% or > 8.5%) and region (North America or Europe). Randomization uses permuted blocks of size 6.

Participants receive study drug for 52 weeks. The primary efficacy assessment is at Week 26; the treatment period continues to Week 52 to characterise durability and longer-term safety.

4.2 Scientific Rationale for Study Design

A placebo comparator on a background of metformin is the accepted design for registration of a new glucose-lowering agent and allows the treatment effect to be estimated without confounding by an active comparator. Rescue medication is allowed for persistent hyperglycemia so that participants are not left with uncontrolled glucose.

4.3 Justification for Dose

The 7 mg and 14 mg doses were selected from the Phase 2 dose-response study, in which 14 mg gave near-maximal HbA1c reduction and doses above 14 mg increased gastrointestinal events without additional efficacy.

4.4 End-of-Study Definition

A participant has completed the study when they complete the Week 56 follow-up visit. The end of the study is the last visit of the last participant.

5 STUDY POPULATION

5.1 Inclusion Criteria

To be eligible, an individual must meet all of the following criteria:

1. Provision of signed and dated informed consent form.
2. Male or female, aged 18 to 70 years inclusive at screening.
3. Diagnosis of type 2 diabetes mellitus for at least 6 months before screening.
4. HbA1c >= 7.5% and <= 9.5% at screening (central laboratory).
5. Treatment with a stable dose of metformin >= 1500 mg/day for at least 12 weeks before screening.
6. Body mass index (BMI) >= 25.0 and <= 40.0 kg/m2 at screening.
7. Estimated glomerular filtration rate (eGFR) >= 60 mL/min/1.73m2 (CKD-EPI 2021) at screening.
8. Able to read and speak English and to complete study questionnaires without assistance.
9. Women of childbearing potential must agree to use a highly effective method of contraception for the duration of the study and for 30 days after the last dose.
10. Willing and able to comply with all study procedures, including daily SMBG and the electronic diary.

5.2 Exclusion Criteria

An individual who meets any of the following criteria will be excluded:

1. Type 1 diabetes, latent autoimmune diabetes in adults, or diabetes secondary to pancreatitis or pancreatectomy.
2. Treatment with any glucose-lowering agent other than metformin within 12 weeks before screening, including any prior use of a GLP-1 receptor agonist or insulin at any time.
3. Fasting plasma glucose > 240 mg/dL at screening.
4. Myocardial infarction, stroke, unstable angina, or hospitalisation for heart failure within 12 months before screening.
5. New York Heart Association (NYHA) class II to IV heart failure.
6. Alanine aminotransferase (ALT) or aspartate aminotransferase (AST) > 2.0 x upper limit of normal (ULN) at screening.
7. History of acute or chronic pancreatitis.
8. Personal or family history of medullary thyroid carcinoma or multiple endocrine neoplasia type 2.
9. Proliferative diabetic retinopathy or maculopathy requiring treatment.
10. History of bariatric surgery or planned weight-loss surgery.
11. Systolic blood pressure >= 160 mmHg or diastolic blood pressure >= 100 mmHg at screening.
12. Any malignancy within 5 years, except treated basal cell carcinoma of the skin.
13. Pregnant or breastfeeding, or planning to become pregnant during the study.
14. Participation in another interventional study within 90 days before screening.
15. Any condition that, in the investigator's opinion, would make the participant unsuitable for the study.

5.3 Lifestyle Considerations

Participants will continue their usual diet and physical activity and will receive standardised diabetes self-management counselling at randomization.

5.4 Screen Failures

Screen failures are individuals who consent but are not randomized. Rescreening is allowed once, at least

4 weeks after the screen failure, except for participants excluded for HbA1c, who may not be rescreened.

5.5 Strategies for Recruitment and Retention

Participants will be recruited from site clinic databases and primary care referral networks. Retention will be supported by visit reminders, travel reimbursement and a participant newsletter.

6 STUDY INTERVENTION

6.1 Study Intervention Administration

SYN-4417 is supplied as 7 mg and 14 mg film-coated tablets with matching placebo tablets identical in appearance. Participants take one tablet orally once daily in the morning with or without food. Participants in the 14 mg arm take 7 mg for the first 4 weeks; the escalation is masked with placebo so that all arms follow the same visible regimen.

6.2 Measures to Minimize Bias

Treatment assignment is concealed through the IWRS. Investigators, participants and the sponsor study team remain blinded until database lock. Emergency unblinding is available through the IWRS when knowledge of the treatment is needed for the participant's medical care.

6.3 Study Intervention Compliance

Compliance is assessed by tablet counts at each dispensing visit. Participants taking fewer than 80% of expected doses between two visits will be counselled.

6.4 Concomitant Therapy and Rescue Medication

Metformin must be continued at the screening dose throughout the study. Rescue medication (any approved agent except GLP-1 receptor agonists and DPP-4 inhibitors) may be started if fasting plasma glucose is above 270 mg/dL from Week 0 to Week 12, above 240 mg/dL from Week 12 to Week 26, or HbA1c is above 8.5% after Week 26, confirmed on repeat testing.

7 DISCONTINUATION AND WITHDRAWAL

Participants may discontinue study drug for adverse events, pregnancy, confirmed pancreatitis, or at the investigator's discretion. Participants who stop study drug are asked to remain in the study and attend all remaining visits. Participants may withdraw consent at any time without giving a reason.

8 STUDY ASSESSMENTS AND PROCEDURES

8.1 Efficacy Assessments

HbA1c, fasting plasma glucose and body weight are measured at the visits shown in the Schedule of Activities. All laboratory samples are analysed by a central laboratory. Body weight is measured in light clothing without shoes on a calibrated scale.

8.2 Safety Assessments

Safety assessments include adverse events, vital signs, physical examination, 12-lead ECG, clinical laboratory tests and hypoglycemic events recorded in the electronic diary. Participants record SMBG values at least twice weekly and whenever they have symptoms of hypoglycemia.

8.3 Adverse Events and Serious Adverse Events

An adverse event (AE) is any untoward medical occurrence in a participant, whether or not related to study drug. Serious adverse events (SAEs) must be reported to the sponsor within 24 hours of the site becoming aware of them. AEs are collected from randomization until the Week 56 follow-up visit.

8.4 Hypoglycemia

Hypoglycemia is classified according to the American Diabetes Association: level 1 (glucose < 70 and >= 54 mg/dL), level 2 (glucose < 54 mg/dL), and level 3 (severe event requiring assistance).

8.5 Adverse Events of Special Interest

Adverse events of special interest are acute pancreatitis, gallbladder events, thyroid neoplasms, severe gastrointestinal events and diabetic retinopathy complications. Suspected pancreatitis events are adjudicated by an external adjudication committee.

9 STATISTICAL CONSIDERATIONS

9.1 Statistical Hypotheses

The primary null hypothesis is that the mean change from baseline in HbA1c at Week 26 is equal for SYN-4417 14 mg and placebo. The 7 mg comparison is tested next in a fixed sequence, followed by the key secondary endpoints, to control the overall type I error at 0.05.

9.2 Sample Size Determination

A total of 600 participants (200 per arm) will be randomized. With 200 participants per arm, the study has 90% power to detect a difference of 0.3% in mean HbA1c change from baseline at Week 26 between SYN-4417 14 mg and placebo, assuming a common standard deviation of 1.1% and a two-sided significance level of 0.05.

9.3 Populations for Analyses

The full analysis set (FAS) includes all randomized participants who take at least one dose of study drug. The safety set includes all participants who take at least one dose, analysed by treatment received.

9.4 Statistical Analyses

The primary endpoint is analysed with a mixed model for repeated measures (MMRM) with treatment, visit, stratification factors and treatment-by-visit interaction as fixed effects and baseline HbA1c as a covariate. Data collected after initiation of rescue medication are included. Missing data are assumed to be missing at random.

Binary endpoints are analysed with logistic regression adjusted for the stratification factors. Safety data are summarised descriptively.

9.5 Interim Analyses

No interim analysis for efficacy is planned.

10 REGULATORY, ETHICAL AND STUDY OVERSIGHT

10.1 Informed Consent Process

The investigator or a delegate explains the study to each potential participant before any study procedure. The participant is told that the study involves 12 clinic visits over about 14 months, daily study tablets, regular blood tests and a home glucose diary. Consent is documented with the participant's signed and dated consent form, and a copy is given to the participant.

10.2 Confidentiality and Privacy

Participant data are identified by a study number only. Records are stored securely at the site and in the sponsor's validated electronic data capture system in accordance with applicable data protection law.

10.3 Safety Oversight

Safety oversight will be performed by the sponsor's medical monitor, who reviews blinded safety data on an ongoing basis. Individual safety findings may be escalated to the sponsor's internal safety review team.

10.4 Clinical Monitoring

Sites are monitored through a combination of on-site and remote visits following a risk-based monitoring plan.

10.5 Quality Assurance and Quality Control

The sponsor or its delegate may audit any site. Sites must allow direct access to source data and documents for monitoring, audits and regulatory inspections.

10.6 Publication and Data Sharing

The study will be registered in a public trial registry before the first participant is enrolled, and results will be published regardless of outcome.

10.7 Conflict of Interest

Any investigator with a financial interest in the sponsor must disclose it before taking part in the study.

11 REFERENCES

1. American Diabetes Association Professional Practice Committee. Standards of Care in Diabetes. Diabetes Care, current edition.

2. ICH E9(R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials, 2019.

3. ICH M11 Clinical electronic Structured Harmonised Protocol (CeSHarP) guideline.

4. Synthetic Therapeutics. Clinical Study Report SYN-T2D-201 (fictional), 2025.

12 ABBREVIATIONS

AE: adverse event

ALT: alanine aminotransferase

AST: aspartate aminotransferase

BMI: body mass index

eGFR: estimated glomerular filtration rate

FAS: full analysis set

GLP-1: glucagon-like peptide-1

HbA1c: glycated haemoglobin

IWRS: interactive web response system

MMRM: mixed model for repeated measures

SAE: serious adverse event

SMBG: self-monitoring of blood glucose

T2DM: type 2 diabetes mellitus

ULN: upper limit of normal